

```
#include <ESP32Servo.h>
#define IR_LEFT 34
#define IR_RIGHT 35
#define IN1 26
#define IN2 27
#define IN3 14
#define IN4 23
#define EN1 25
#define EN2 33
#define TRIG 5
#define ECHO 18
#define SERVO_PIN 4
```

```
Servo myServo;
```

```
int threshold = 20;
int lastTurn = 0;
```

```
void enableMotors() {
    digitalWrite(EN1, HIGH);
    digitalWrite(EN2, HIGH);
}
```

```
void disableMotors() {
    digitalWrite(EN1, LOW);
    digitalWrite(EN2, LOW);
}
```

```
void moveForward() {
    enableMotors();
    digitalWrite(IN1, HIGH);
    digitalWrite(IN2, LOW);
    digitalWrite(IN3, HIGH);
    digitalWrite(IN4, LOW);
}
```

```
void stopMotors() {
    disableMotors();
}
```

```
void turnLeft() {
    enableMotors();
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, LOW);
    digitalWrite(IN3, HIGH);
    digitalWrite(IN4, LOW);
}
```

```
void turnRight() {
    enableMotors();
    digitalWrite(IN3, LOW);
}
```

```

digitalWrite(IN4, LOW);
digitalWrite(IN1, HIGH);
digitalWrite(IN2, LOW);
}

void moveBackward() {
  enableMotors();
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
}

long getDistance() {
  digitalWrite(TRIG, LOW);
  delayMicroseconds(2);

  digitalWrite(TRIG, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG, LOW);

  long duration = pulseIn(ECHO, HIGH, 30000);
  long distance = duration * 0.034 / 2;

  if (distance == 0) distance = 100;
  return distance;
}

int scanFront() {
  myServo.write(90);
  delay(200);
  return getDistance();
}

int scanLeft() {
  myServo.write(150);
  delay(300);
  return getDistance();
}

int scanRight() {
  myServo.write(30);
  delay(300);
  return getDistance();
}

void followLine() {
  int leftIR = digitalRead(IR_LEFT);
  int rightIR = digitalRead(IR_RIGHT);

  if (leftIR == LOW && rightIR == LOW) {

```

```

    moveForward();
}
else if (leftIR == HIGH && rightIR == LOW) {
    turnLeft();
}
else if (leftIR == LOW && rightIR == HIGH) {
    turnRight();
}
else {
    stopMotors();
}
}

```

```

void avoidObstacle() {
    stopMotors();
    delay(150);

```

```

    int left = scanLeft();
    int right = scanRight();

```

```

    myServo.write(90);
    delay(100);

```

```

    if (left > right && left > threshold) {
        turnLeft();
        delay(500);
        lastTurn = -1;
    }
    else if (right > left && right > threshold) {
        turnRight();
        delay(500);
        lastTurn = 1;
    }
    else {
        moveBackward();
        delay(500);

```

```

        if (lastTurn == -1) {
            turnRight();
            lastTurn = 1;
        } else {
            turnLeft();
            lastTurn = -1;
        }
        delay(600);
    }
}

```

```

void setup() {
    pinMode(IR_LEFT, INPUT);
    pinMode(IR_RIGHT, INPUT);

```

```
pinMode(IN1, OUTPUT);  
pinMode(IN2, OUTPUT);  
pinMode(IN3, OUTPUT);  
pinMode(IN4, OUTPUT);
```

```
pinMode(EN1, OUTPUT);  
pinMode(EN2, OUTPUT);
```

```
pinMode(TRIG, OUTPUT);  
pinMode(ECHO, INPUT);
```

```
myServo.attach(SERVO_PIN);  
myServo.write(90);
```

```
Serial.begin(115200);  
}
```

```
void loop() {
```

```
    int front = scanFront();  
    Serial.println(front);
```

```
    if (front < threshold) {  
        avoidObstacle();  
    } else {  
        followLine();  
    }  
}
```